AIRSTREET · CALCULATION

AirStreet sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-24	Amish Chadha	STEM check fixes made under Amish's standing instruction of 2026-10-23 (AST-DDR-005): NO2 noise from the datasheet (C2), ozone filter life (C7), rated-range flags (H5), band thickness read from the model; hollow shield spacers of the D3 model (3.77 kg with the sun shield); requirements cited from AST-PRS-001
0.6	2026-10-02	Amish Chadha	sizing.py re-run with R13 at 4.0 kg including the sun shield and the four lightening steps in the model (40 x 4 mm rail, plate windows, 1.5 mm shield plates, saddles at 40 % infill); lines 3 and 8 repriced; R13 met on paper; cost in the approved value-engineering wording
0.5	2026-10-02	Amish Chadha	R13 status against the 4.0 kg limit accepted on 2026-10-02 (AST-DDR-003 A1); sizing.py still to be re-run with it
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target

0.3	2026-09-30	Amish Chadha	Design made constructable (AST-DDR-003); mass, area, mounting loads and cost recalculated; R13 and R15 now not met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (power and energy, airtime and storage, NO2 error budget, radiation shield, PM2.5 humidity error, geometry, mass and area, wind and mounting, service, cost)

AirStreet sizing calculations

Version 0.7 (2026-10-24) adds the STEM check fixes of AST-DDR-005, made under Amish's standing instruction of 2026-10-23: the NO2 noise term now uses the sensor datasheet's 15 ppb at 2 sigma (C2), the NO2 cell is replaced before its ozone filter is used up (C7), every record carries flags for readings outside the sensors' rated ranges (H5), and the band thickness is read from the model (0.6 mm, as the mass always used); the D3 model's hollow shield spacers take 0.01 kg off. The independent review of 2026-10-24 (AST-STM-001) then corrected the NO2 budget (the noise averages over 12 independent records an hour, not 120 samples; a 5 % slope term; the EU reference temperature of 20 °C), the cross arm bolt force, the rail check (twist and minimum yield) and the ozone rule, and added sensitivity lines (C8, D4, B9, G1, G8). The hourly NO2 error rises from about 3.9 to 4.3 ppb and the annual uncertainty from 4.25 to 4.65 µg/m³; no requirement status changes. Version 0.6 followed the constructable design of AST-DDR-003 (2026-09-30), which made every part of the node buildable and moved onto FieldNode's own constructable core, with the four lightening steps of AST-DDR-003, A1 (accepted on 2026-10-02) now in the model: a 40 x 4 mm rail, windows in the adapter plates, 1.5 mm shield plates and saddles printed at 40 % infill. On paper AirStreet now meets nine of its sixteen requirements (five by calculation, four by design), has five at risk and is over its cost target on one (R15); R3 stays withdrawn. The node weighs 3.61 kg on the pole and 3.77 kg with FieldNode's sun shield, against R13's limit of 4.0 kg including the shield (set on 2026-10-02; it was 3.5 kg), so R13 is met on paper with 0.23 kg of margin; before the lightening it was 4.00 kg, on the limit. The pole owner of the first site is still to confirm the 4.0 kg load. Value-engineering target: USD 285. Estimated cost of the constructable design: USD 302.50 (USD 17.50 over the target); the cost drivers and savings worth trying are in the design decisions register. The frontal area, 0.139 m², stays inside 0.15 m², and every mounting part carries its load with a large margin. Power and airtime are unchanged from v0.2, the NO2 error budget is larger after the review (4.3 ppb hourly MAE, 4.65 µg/m³ annual), and the thinner shield plates change the shield and humidity results only in the second decimal: the 60 s particle run gives an 87.0 mW design sensor load inside FieldNode's 100 mW allowance and 6.6 days without sun, and an annual NO2 mean carries about 4.65 µg/m³ of uncertainty (1σ). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

> **Safety:** These are first-principles estimates for a paper proof of concept. They do not show that the node is safe to mount on any particular pole, and they are not a substitute for the pole owner's structural check, for electrical checks on the FieldNode cell or for field collocation of the sensors. AirStreet data are indicative, not a regulatory measurement or health advice. See AST-PRC-001, Safety.

Scope and method

The note checks every requirement in AST-PRS-001 v0.9 (cited here by the R numbers in its "Was" column: R1 is PR-001, and so on) against the design in AST-PRC-001 v0.9 and the parametric model `cad/src/model.py`, which builds the FieldNode core from FieldNode's own model (vendored as `cad/src/fieldnode_core.py`). The script imports the model's `PARAMS`, derived dimensions and part solids, so the pod, shield, rail, clamp positions and projected areas used here are those in the STEP files and in drawing AST-DWG-001. It reads `bom/bom.csv` and `budget_usd` in `project.yaml`. FieldNode figures are taken from FND-CAL-001 (v0.1 for power and energy, v0.3 for mass) in the FieldNode repo. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

The design case is a 140 mm street light pole with the inlet plane 3.0 m above the sidewalk (AST-DDR-001 D7), one record every 5 min with a 60 s PM run (D8 and DDR-002), LoRaWAN at SF9 in EU868 through a private gateway (15 min on The Things Network), FieldNode's back plate left off (DDR-002), and ambient -10 to 50 °C.

Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
PM sensor	SPS30 at 5 V, 55 mA typical and 65 mA maximum in measurement mode; 41 x 41 x 12 mm, 26.3 g; up to 30 s to first reading at low particle counts; recommended 10 to 40 °C and 20 to 80 % RH	Sensirion SPS30 datasheet (07/2023), checked 2026-09-25
NO2 chain	Front end and ADC held on continuously, 1.5 mA at 5 V; overall gain 0.25 mV/ppb; 7.5 ppb (1σ) noise per 1 s sample; 10 samples per record; ozone filter 500 ppm h; rated -30 to 40 °C and 15 to 85 % RH continuous	Noise, filter and range from the NO2-B43F data sheet (Alphasense, checked 2026-10-24; 15 ppb at 2σ); current and gain assumed
T and RH	SHT45, ±0.1 °C and ±1 % RH typical	Adafruit SHT45 product page, checked 2026-09-25
Margin	1.2 on the port load, applied to the maximum SPS30 current	Judgment
FieldNode core	Rails 90 % efficient; SX1262 +14 dBm at 45 mA, receive 4.6 mA, two 0.1 s windows, controller 0.5 s at 8 mA per report, 33 µA sleep; 6 Ah LiFePO4 cell, 80 % usable; capacity ratios 0.70 at -20 °C and 0.80 at end of life; 7.75 Wh/day stored in the worst month, 0.8 Wh/day on a hot clear day without a sun shield; 2.41 kg; $126.00	FND-CAL-001 v0.1
Extra controller time	1 s at 8 mA per record to run and read the sensors	Assumed
NO2 calibration	Field-calibrated residual MAE 3.5 ppb, taken as σ = 4.39 ppb; zero drift 0.5 ppb/month, half removed by the anchor node; 2 ppb bias moving from the reference site to a street; 6 h correlation time	Residual from Zimmerman et al., 2018; the rest assumed
PM humidity growth	κ-Köhler volume growth 1 + κ·aw/(1 − aw) with κ = 0.3 ± 0.1; residual after the CalRig transfer check 2 µg/m³ + 5 %	Assumed typical urban aerosol
Shield	1,000 W/m² sun, ground albedo 0.2, absorptance 0.25 (aged white ASA), 15 W/m²K per plate face, view factors from the probe of 0.1 (cap), 0.1 (bottom plate) and 0.6 (other plates, 0.5 K warm); inner air speed 10 to 30 % of a 1 m/s wind	Handbook ranges; assumed
Mass	ASA 1.07, aluminum 2.70, stainless 7.90 g/cm³ from model volumes; FieldNode V-blocks and small clamps (0.20 kg) and back plate (0.47 kg) left off	Estimates
Wind	35 m/s gust; air 1.225 kg/m³; drag coefficient 1.2 on the projected area; band preload 1,000 N (as FND-CAL-001), friction 0.3	Screening values, not a code check

A. Power and energy (R9)

- **PM sensor.** 0.275 W while running at 20 % duty (60 s in 300 s) is 55.0 mW typical, or 65.0 mW at the datasheet maximum [A1].
- **Port load.** The NO2 front end and ADC add 7.5 mW continuously and the T and RH sensor 0.01 mW, for 62.5 mW typical [A2]. With the maximum SPS30 current and the 1.2 margin, the design load is 87.0 mW, against

FieldNode's 100 mW design value and 115 mW ceiling [A3]. R9 is met on paper under either figure, with 13 mW left under the design value.

- **Core.** At a 5 min interval each uplink costs 52.9 mJ at SF9, and the core uses 9.0 mWh/day including sensor wake-ups and sleep, against 4.0 mWh/day for FieldNode alone at 15 min [A4].
- **Daily draw.** 2.33 Wh/day from the cell, a 97.1 mW average [A5].
- **Autonomy.** A full cell lasts 6.6 days without sun, 4.6 days at -20 °C, 5.3 days at end of cell life and 3.7 days cold at end of life [A6]. A LiFePO4 cell is not charged below 0 °C, so a sub-zero spell is ridden through on the stored charge (AST-STM-001 S4). The cold figure is below the 5 days FieldNode's own R6 asks of its core; AirStreet's R9 does not set an autonomy target.
- **Energy balance.** In FieldNode's worst month the cell stores 7.75 Wh/day against 2.33 Wh drawn, 3.3 times the demand [A7]. On a hot clear day without FieldNode's sun shield only 0.8 Wh is stored, a 1.53 Wh/day deficit, which a full cell covers for 10 such days in a row [A8]; the shield is required above 45 °C (DDR-002).
- **Clean air.** The SPS30 needs up to 30 s to its first reading at low particle counts, so the 60 s run (DDR-002) leaves at least 30 s of valid readings. The 30 s run of v0.1 gave 48.0 mW [A9].

B. Data, airtime and storage (R8, R10)

- **Record.** Eight 2-byte values (PM1, PM2.5, PM10, WE, AE, T, RH, status), a 2-byte counter and 2 bytes of flags: 20 bytes, 288 records a day [B1].
- **Airtime.** With 13 bytes of LoRaWAN overhead on a 125 kHz channel, one uplink takes 72 ms at SF7, 134 ms at SF8, 247 ms at SF9, 453 ms at SF10 and 1,810 ms at SF12 [B2]. The TRL 2 figure of about 1.3 s at SF12 was low.
- **EU868 duty cycle.** The worst case, SF12, uses 0.60 % of each 5 min, so the 1 % sub-band limit is met at every spreading factor [B3]. R10 is met on paper.
- **Fair use.** On The Things Network's 30 s/day fair-use policy, a 5 min interval fits only at SF7 (20.7 s); SF8 needs 38.5 s and SF9 71.1 s [B4]. Under DDR-002 the node reports every 5 min only through a private gateway (TwinKit by default, FND-DDR-001 D7), where this limit does not apply, and every 15 min on TTN. At 15 min an uplink takes 12.8 s/day at SF8 and 23.7 s/day at SF9, inside 30 s, but 43.5 s/day at SF10, so fair use is met up to SF9 [B8].
- **US915.** The 400 ms dwell limit allows SF9 (247 ms) and not SF10 (453 ms) [B5].
- **Storage.** 7 days of store and forward take 40.3 kB of FieldNode's 16 MB flash [B6], and an hourly mean is ready within one record interval of the hour's end, at most 15 min on TTN [B7]. A full 7-day backlog of 2,016 records takes about 14 h to send at SF9 within the 1 % duty cycle through a private gateway; on TTN a backlog cannot be flushed within 30 s a day, so store and forward relies on the private gateway [B9]. R8 is met on paper.

C. NO2 error budget (R2, R3, R4)

- **Units and resolution.** 1 ppb = 1.880 µg/m³ at 25 °C; one ADC step of 62.5 µV is 0.25 ppb at the assumed gain [C1].
- **Hourly error.** The hourly terms (1σ) are noise 2.17 ppb (7.5 ppb per sample from the data sheet, averaged over the 12 independent records of an hour: the 10 samples of one record fall within the cell's response time, so they are not independent; v0.6 used 5 ppb over 120 samples, 0.46 ppb), calibration residual 4.39 ppb, drift 0.87 ppb, transfer bias 2.00 ppb and a slope term of 1.00 ppb (a 5 % span error left after a 14-day collocation, at a 20 ppb street mean; assumed), 5.45 ppb in all [C2]. On The Things Network each 15 min uplink carries the mean of its

three 5 min records, so the same 12 records an hour are averaged. That is a mean absolute error of 4.3 ppb (8.2 µg/m³) against the 5 ppb of R2 [C3]. R2 is at risk: every term except noise rests on literature or an assumption (the 3.5 ppb residual comes from a model that also used other gas sensors on its node, so it is optimistic for an NO2 cell alone), and only collocation can show it.

- **Annual mean.** Random terms average out over a year, but the transfer bias, the drift residual, the collocation intercept (0.59 ppb from 14 days) and the slope term do not: 2.47 ppb, or 4.65 µg/m³, against 2 µg/m³ [C4]. With the slope term, 2 µg/m³ cannot be met even with no transfer bias at all [C6]. R3 is therefore withdrawn as a requirement and kept as a research question (DDR-002).
- **EU limit.** The directive refers gas concentrations to 20 °C, where 1 ppb is 1.913 µg/m³. At 90 % one-sided confidence, a street's annual mean can be classified against 40 µg/m³ only if it is below 33.9 or above 46.1 µg/m³ [C5]. R4 is at risk: it holds for streets well clear of the limit but not for those near it. The recast Directive (EU) 2024/2881 lowers the limit to 20 µg/m³ from 2030; against it the band is 13.9 to 26.1 µg/m³ (AST-DEC-001, open decision 4). If the street transfer bias is 3 ppb rather than 2 (a 5 to 10 % fall in NO2 from 1.5 to 3 m), the 40 µg/m³ band widens to 31.8 to 48.2 µg/m³ [C8].
- **Ozone filter.** The cell's ozone filter holds more than 500 ppm h. At a 30 ppb annual mean ozone it lasts 23 months, at 40 ppb 17 months, and at the 50 to 60 ppb of hot, sunny sites 14 to 11 months, against the cell's 24-month life [C7]. So the cell is replaced every 12 months at the pod exchange, or at 18 or 24 months where the collocated reference station's ozone record shows the dose since fitting stays under 500 ppm h by then (AST-DDR-005 F3).

D. Radiation shield (R5)

- **Plate temperatures.** In 1,000 W/m² sun the cap runs 8.3 K and the bottom plate 1.7 K above ambient; the mean radiant excess seen by the probe is 1.30 K [D2].
- **Probe error.** With inside air at 30 % of a 1 m/s wind the probe reads 0.26 K high from radiation and 0.16 K from warmed air, 0.42 K in all; at 10 % it reads 0.38 K and 0.49 K, 0.87 K in all [D1]. The 1.5 mm plates (2 mm before v0.6) leave slightly wider gaps between plates, so a little more air passes. Both are under the 1 °C of R5, but the result depends on the inner air speed, which is assumed: it passes 1 K once the inner air falls below 8.3 % of the wind, so in calm sun the error exceeds 1 K [D4]. The result is fragile and is checked against an aspirated reference at TRL 4.
- **Humidity.** A 0.87 K warm error reads 2.4 % RH low at 50 % RH and 4.1 % RH low at 85 % RH, at 30 °C [D3]. The SHT45 alone meets ±0.5 °C and ±3 % RH, but the shield error at high humidity does not. R5 is at risk.

E. PM2.5 error with humidity (R1)

- **RH input.** The humidity correction uses RH with an uncertainty of 4.3 % (1 % sensor and the shield error at 85 % RH) [E1].
- **Error against the bound.** Table 2 compares the combined error with the R1 bound of ±5 µg/m³ or ±30 % [E2].

Table 2. Hourly PM2.5 error after humidity correction, µg/m³ (bound in parentheses).

RH	10 µg/m³	25 µg/m³	50 µg/m³
50 %	2.2 (5.0)	3.2 (7.5)	5.4 (15.0)
70 %	2.6 (5.0)	4.7 (7.5)	8.6 (15.0)
80 %	3.1 (5.0)	6.3 (7.5)	12.1 (15.0)
85 %	3.6 (5.0)	**7.8 (7.5)**	**15.2 (15.0)**
90 %	4.7 (5.0)	**10.8 (7.5)**	**21.4 (15.0)**

- **Result.** R1 is met on paper up to 80 % RH at every level tested, which is also the top of the SPS30's recommended humidity range [E3]. Above 85 % RH the particles' water uptake is steep and the correction is uncertain. R1 is at risk for humid climates and nights.

F. Geometry, area and mass (R11, R13)

- **Fit.** Inlet plane 3,000 mm above the sidewalk, inside the EU 1,500 to 4,000 mm range; 140° V-saddles and bands cut to length fit 80 to 200 mm poles; no drilling [F1]. R11 is met by design.
- **Layout.** FieldNode's whip clears the pod's drip lid by 15 mm (23 mm in v0.2, before FieldNode's antenna moved 28 mm toward the pod and the pod moved 20 mm away), the clamps are 375 mm apart, the node is 706 mm tall and it stands 172 mm out from the pole face (173 mm before the rail was thinned) [F2].
- **Area.** Rasterizing the model gives 0.139 m² seen from the street and 0.054 m² along it, against 0.15 m² [F3] (0.143 m² in v0.5, 0.126 m² in v0.2). FieldNode's constructable bracket, the adapter plates and the cross arm add to the outline; the plate windows above and below the enclosure take a little off it. R13's area limit is met on paper.
- **Mass.** FieldNode core as used 1.89 kg (FieldNode's base node of 2.45 kg less its back plate, V-blocks and bands, 0.56 kg); rail 0.19 kg; adapter plates 0.21 kg; cross arm 0.19 kg; V-saddles 0.10 kg (6 walls and 40 % infill, 57 % of solid); pod shell and floor 0.37 kg; shield plates, cap and spacers 0.22 kg (the spacers are tubes with a 5.5 mm bore for the rods since v0.7); bands 0.10 kg (12.7 x 0.6 mm, read from the model); shield rods 0.06 kg; sensors, front end, probe, cables, glands, inserts and fixings 0.29 kg [F4]. In all 3.61 kg on the pole and 3.77 kg with FieldNode's sun shield at a hot site, against 4.0 kg including the shield: 0.23 kg of margin. The sensor head and mount are 1.72 kg [F5]. R13 is met on paper; the pole owner of the first site is still to confirm the load.
- **Lightening.** The four steps of AST-DDR-003, A1 took 0.22 kg off (3.84 to 3.62 kg without the sun shield and 4.00 to 3.78 kg with it); with the hollow spacers of v0.7 the totals are 3.61 and 3.77 kg [F8]. The saddles save 0.08 kg, the thinner rail 0.05 kg, the plate windows 0.06 kg and the thinner shield plates 0.04 kg. The "about 3.6 kg" of AST-DDR-003, A1 is the node without the sun shield; with it the lightened node is 3.77 kg, close to the 3.76 kg estimated when the recommendation was reviewed.
- **Why it rose.** FieldNode's constructable core is 0.15 kg heavier as AirStreet uses it, and the parts that make AirStreet buildable (adapter plates in place of adapter bars, the cross arm in place of the short shield arm, larger saddles, the separate pod floor, spacers and fixings) add the rest [F6]. Option (c) of AST-DDR-003, A1 was accepted on 2026-10-02: the limit is 4.0 kg and the four lightening steps are now in the design.
- **Bands.** Each band runs 487 mm round the 140 mm design pole and its saddle, 326 mm round an 80 mm pole and 664 mm round a 200 mm pole, plus about 100 mm for the housing and tail [F7].

G. Wind and mounting

- **Load.** A 35 m/s gust gives 750 Pa and 125 N on the node at about 3.33 m, or 416 N·m at the pole base [G1]. 35 m/s is a screening speed; where the local code asks 45 m/s the load is 206 N and 688 N·m. The pole's capacity is for its owner to confirm.
- **Clamps.** The downward load of 75 N (weight and wind on the tilted panel) compares with 1,885 N of friction from two bands, a factor of 25 [G2]. Twist about the pole from wind on the offset shield is 2.42 N·m against 132 N·m [G5]. Both rest on the assumed 1,000 N band preload.
- **Cross arm.** The 30 x 30 x 3 mm angle carries the shield (0.30 kg, 12.1 N of wind) 200 mm to the right of the rail and the pod (0.46 kg) 90 mm to the left: 0.58 N·m vertical and 2.42 N·m horizontal at the rail, 4.4 MPa [G3]. Its two M5 bolts, 20 mm apart, resist the wind's twist as a couple: the worse bolt takes about 127 N in the gust, against about 7,000 N proof load [G4] (67 N before the review, which halved the couple).
- **Rail and pod.** The rail at the lower clamp carries 26.6 N and 2.73 N·m, 25.6 MPa in the 40 x 4 mm bar (16.4 MPa in the 40 x 5 mm bar of v0.5) [G6]. Each of the pod's two hanging screws takes about 11 N in the gust, far below what an M5 heat-set insert in ASA holds; the insert pull-out is to be checked at TRL 4 [G7]. With the cross arm's 2.42 N·m twist the rail also carries 12.2 MPa of torsion, 33.2 MPa von Mises, against 73 MPa (the minimum yield of 6063-T5, about 110 MPa, over 1.5) [G8]. Every stress is well below yield.

H. Service and environment (R12, R16)

- **Service time.** Under DDR-002 the pod is exchanged whole: ladder 5 min, unplug two M12 plugs and the probe lead 1.5 min, release the pod 1.5 min, fit the pre-collocated pod 2 min, reconnect and check glands 1.5 min, uplink check 2 min: about 14 min against 15 min [H1]. The exchange pod arrives with its collocation and CalRig record, and the returned pod is serviced and recollocated at the reference site [H2]. R16 is met on paper, with a small margin.
- **Temperature range.** FieldNode is rated for -20 to +45 °C ambient and reaches 58.7 to 73.3 °C inside at 45 °C, while AirStreet's R12 asks for -10 to 50 °C [H3]. The SPS30's recommended range is 10 to 40 °C and 20 to 80 % RH, with absolute limits of -10 to 60 °C and 0 to 95 % RH [H4]. Under DDR-002 FieldNode's sun shield is required at sites above 45 °C, so R12 asks for it there; FieldNode has now designed it (FND-DDR-003), and it fixes to AirStreet's adapter plates (AST-DDR-003). R12 stays at risk; condensation and insects in the pod remain unverified.
- **Rated ranges.** The NO2-B43F is rated -30 to 40 °C and 15 to 85 % RH continuous, and the SPS30 is recommended for 10 to 40 °C and 20 to 80 % RH, while R12 asks the node to survive -10 to 50 °C [H5]. Every record therefore carries flags (AST-DDR-005 F2): shield air above 35 °C (40 °C less an estimated 5 K by which the sunlit pod air runs above the shield air, to measure at TRL 4; 3 K before the review), RH above 80 % for PM and above 85 % for NO2, RH below 20 % for PM and below 15 % for NO2, and below 10 °C for PM. Flagged hours are kept and published with the flag, not dropped, so a map can show them; the flag is set from the shield's T and RH already in the record, so the record stays 20 bytes. R7 now names the flags; R12 stays at risk.

I. Cost (R15)

- **Totals.** The BOM has 11 lines, all priced: $433.50 per node, of which the FieldNode core is $131.00 (FieldNode's $139.00 less its unused $8.00 pole mounting kit; costed in FieldNode, AST-DDR-001 D1) and the sensor head $302.50 [I1].
- **Against the value-engineering target.** Value-engineering target: USD 285. Estimated cost of the constructable design: USD 302.50 (USD 17.50 over the target) [I2]. `budget_usd` is a hypothetical control target. **R15 is over**

the target by $17.50. The rise from $285.00 is in line 3 (band stock, adapter plates and cross arm, +$4.50), line 9 (probe tube, +$1), line 10 (probe lead with its M8 plug, +$4) and line 11 (inserts, glands, probe socket, terminal block and fixings, +$8). Line 8 is back to $9.00: the thinner shield plates save about as much filament as the spacers use. In line 3 the thinner rail saves about $0.50, and the saddles are repriced at about $3.00 of filament at 40 % infill (the $1.50 used before was low), so line 3 is $21.50. The cost drivers and savings worth trying are in the design decisions register and AST-DDR-003, A2.

- **Running costs.** A replacement NO2 sensor every 12 months (18 or 24 where the site's ozone record allows) [C7], and the collocation time, are not in the BOM [I3].

J. Results

Table 3. Requirement status (AST-PRS-001 v0.9, by the R numbers in its "Was" column). Counts, as the script's line [J1] prints them: 1 over the value-engineering target, 5 at risk, 5 met on paper, 4 met by design, 1 withdrawn.

ID	Requirement	Value	Target	Status
R15	Low cost	Sensor head $302.50; $433.50 with FieldNode [I1]	Sensor head $285 value-engineering target (`budget_usd`)	Over the target by $17.50
R1	PM2.5	Within the bound up to 80 % RH, outside it above 85 % RH [E2]	±5 µg/m³ or ±30 % hourly	At risk
R2	NO2	Hourly MAE about 4.3 ppb [C3]	5 ppb or less	At risk
R4	EU limit classification	Decisive outside 33.9 to 46.1 µg/m³ [C5]	90 % confidence at 40 µg/m³	At risk
R5	T and RH	Shield 0.42 to 0.87 K at 1 m/s; 4.1 % RH low at 85 % RH [D1, D3]	±0.5 °C, ±3 % RH; radiation 1 °C or less	At risk
R12	Survive outdoors	FieldNode sun shield required above 45 °C, designed and fits the adapter plates; SPS30 recommended to 80 % RH [H3, H4]	-10 to 50 °C; shield above 45 °C	At risk
R13	Light and compact	3.77 kg with the sun shield (3.61 kg without); 0.139 m² [F3, F5]	4.0 kg with the sun shield (3.5 kg until 2026-10-02); 0.15 m²	Met on paper (0.23 kg margin)
R8	Reporting	5 min on the private gateway; 15 min on TTN, fair use met up to SF9; 7 days = 40.3 kB [B6, B8]	5 min private, 15 min TTN; hourly within 15 min; 7 days	Met on paper
R9	Power	87.0 mW design load; 2.33 Wh/day against 7.75 Wh [A3, A7]	115 mW or less; energy neutral	Met on paper
R10	Duty cycle	0.082 % at SF9, 0.60 % at SF12 [B3]	Below 1 %	Met on paper
R16	Serviceable	About 14 min for a pod exchange [H1]	Pre-collocated pod in 15 min	Met on paper
R6	Stated calibration	CalRig for T, RH, PM; collocation per D3	Calibration record per node	Met by design
R7	Raw data	Raw WE, AE, PM, T, RH, version tag and rated-range flags [H5]	Raw signals kept	Met by design
R11	Pole mounting	80 to 200 mm, V-saddles and bands, inlet 3.0 m [F1]	No drilling; 1.5 to 4 m	Met by design
R14	Levels only	Levels, T and RH only	No images, audio or identifiers	Met by design
R3	NO2 near the WHO guideline	4.65 µg/m³ annual (1σ) [C4]	Withdrawn; research question	Withdrawn

Table 4. TRL 2 numbers checked (v0.1).

TRL 2 figure	TRL 3 value	Action
Sensor power about 41 mW, 45 mW with margin	35.0 mW typical, 48.0 mW design	Precis updated
Airtime about 1.3 s at SF12, 0.5 %	1.81 s, 0.60 %	Precis and requirements updated
Mass about 3.0 kg	3.65 kg (FieldNode 2.41 kg, not 1.7 kg)	R13 not met in v0.1; see Table 5
Frontal area about 0.09 m²	0.138 m² from the model	R13 not met in v0.1; see Table 5
Cost $391 per node; $265 sensor head	$409.00; $283.00	BOM repriced
NO2 hourly error about 5 ppb	About 3.9 ppb MAE, budget of assumed terms	R2 stays at risk

Table 5. Changes from v0.1 under AST-DDR-002.

Quantity	v0.1	v0.2	Decision
PM run per record	30 s	60 s	PM run length
Design sensor load	48.0 mW	87.0 mW	PM run length
Autonomy without sun	11.9 days	6.6 days	PM run length
Reporting	5 min on any network	5 min private gateway; 15 min on TTN	Fair use on TTN
Mass on the pole	3.65 kg	3.26 kg	Back plate left off
Frontal area	0.138 m² against 0.12 m²	0.126 m² against 0.15 m²	Back plate left off; area limit relaxed
Sensor head cost	$283.00 against $180	$285.00 against $285	Budget figure; adapter bars
Service	14 min sensor swap, recollocation needed	14 min pod exchange	Pod as exchange unit
R3	Not met	Withdrawn, research question	WHO guideline

Table 6. Changes from v0.2 under AST-DDR-003 (design for construction).

Quantity	v0.2	v0.3	Reason
Mass on the pole	3.26 kg	3.84 kg (4.00 kg with the sun shield)	FieldNode's constructable core; adapter plates, cross arm, saddles, pod floor, fixings
Frontal area	0.126 m²	0.143 m²	FieldNode's bracket, plates and cross arm
Wind on the node	114 N, 379 N·m	128 N, 428 N·m	Larger area
Sensor head cost	$285.00	$303.00	Parts added for construction
FieldNode core cost	$126.00	$131.00	FieldNode's constructable BOM less its pole mounting kit
Whip clearance to the pod	23 mm	15 mm	FieldNode's antenna moved; pod moved 20 mm left
Shield arm	20 x 8 mm bar, 4.6 MPa	30 x 30 x 3 mm cross arm carrying pod and shield, 4.5 MPa	Pod and shield needed fixings
R13	Met on paper	Not met against 3.5 kg; at risk against 4.0 kg	Limit set at 4.0 kg with the sun shield, AST-DDR-003 A1, accepted 2026-10-02
R15	Met on paper	Over the value-engineering target by $18.00	Parts added for construction, AST-DDR-003

Table 7. Changes from v0.5 under AST-DDR-003, A1 (lightening steps, accepted 2026-10-02).

Quantity	v0.5	v0.6	Reason
Rail	40 x 5 mm, 0.24 kg, 16.4 MPa	40 x 4 mm, 0.19 kg, 25.6 MPa	Thinner bar
Adapter plates	0.27 kg	0.21 kg	Two windows in each plate
V-saddles	0.18 kg (solid)	0.10 kg (6 walls, 40 % infill)	Print settings counted in the mass
Shield plates, cap and spacers	0.27 kg	0.23 kg	Plates 1.5 mm (was 2 mm)
Mass on the pole	3.84 kg (4.00 kg with the sun shield)	3.62 kg (3.78 kg with the sun shield)	The four steps together
Frontal area	0.143 m²	0.139 m²	Plate windows
Wind on the node	128 N, 428 N·m	125 N, 416 N·m	Smaller area
Shield error at 1 m/s	0.43 to 0.89 K	0.42 to 0.87 K	Wider gaps between thinner plates
Sensor head cost	$303.00	$302.50	Lines 3 and 8 repriced
R13	At risk (4.00 kg against 4.0 kg)	Met on paper (3.78 kg against 4.0 kg)	Lightening steps in the design
R15	Over the target by $18.00	Over the target by $17.50	Lines 3 and 8 repriced

Table 8. Changes from v0.6 under AST-DDR-005 (STEM check fixes, 2026-10-24, made under Amish's standing instruction of 2026-10-23).

Quantity	v0.6	v0.7	Reason
NO2 noise per sample	5 ppb (1σ), assumed, over 120 samples an hour	7.5 ppb (1σ), data sheet 15 ppb at 2σ, over 12 independent records	F1 and the review
NO2 slope term	None	1.00 ppb (5 % at 20 ppb, assumed)	Review
NO2 hourly total	4.92 ppb, MAE 3.9 ppb	5.45 ppb, MAE 4.3 ppb	F1 and the review
NO2 annual (1σ)	2.26 ppb, 4.25 µg/m³	2.47 ppb, 4.65 µg/m³	Review
EU limit band (R4)	34.6 to 45.4 µg/m³ at 25 °C	33.9 to 46.1 µg/m³ at the directive's 20 °C	Review
NO2 cell replacement	About every 2 years	Every 12 months, or 18 or 24 where the ozone record allows	F3 and the review
Rated-range flags	None	Shield air above 35 °C; RH above 80 % (PM) or 85 % (NO2), below 20 % (PM) or 15 % (NO2); below 10 °C (PM)	F2 and the review
Cross arm bolt force	67 N	127 N (couple on one bolt)	Review
Rail check	25.6 MPa bending against about 150 MPa	33.2 MPa von Mises with twist against 73 MPa (minimum yield over 1.5)	Review
Band in the model	12.7 x 0.8 mm (calculation used 0.6 mm)	12.7 x 0.6 mm, read from the model	F4
Shield plates, cap and spacers	0.229 kg (solid spacers in the model)	0.222 kg (spacers with a 5.5 mm bore)	F5, D3 model
Mass on the pole	3.62 kg (3.78 kg with the sun shield)	3.61 kg (3.77 kg)	F5
Downward load on the clamps	76 N	75 N	F5